

Rahul Rawat

Master's thesis: Agentic Pentesting and AI Agents for Pentest Workflows

P E R S O N A L D E T A I L S

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Date of birth	20 February 2002
Nationality	Indian, student visa with valid work permit
Availability	Working student 20 hours per week immediately, full-time from April 2027

P R O F I L E

Master's student in Data Science and Analytics at SRH Heidelberg, based in Mannheim, with hands-on experience in AI agents, Retrieval Augmented Generation and LLM orchestration. I have built a Hybrid RAG orchestrator with agentic routing over Llama **3.1 8B** via Groq and LangChain, delivered a fairness-by-design credit scoring system with honest evaluation aligned to the EU AI Act, and implemented an end-to-end time-series pipeline at eRay GmbH with gate checks and governance rules. Confident with Python, LangChain, Git and API-based LLM services, with a clear focus on evaluation, guardrails and traceability of agentic workflows.

W O R K E X P E R I E N C E

Oct 2025 – Mar 2026
Heidelberg

eRay GmbH

Data Scientist

- In a **6-month** collaboration between eRay GmbH and SRH University Heidelberg to forecast lake water quality, delivered an end-to-end recursive time-series pipeline over dissolved oxygen, turbidity and pH across a **40-feature** space with a target lag set of lag_1h, lag_24h, lag_3d, lag_7d, lag_roll_mean_24h and lag_roll_std_24h.
- Directly compared **6 candidates** — Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost and Prophet — with strict tree constraints (max_depth 4, learning_rate 0.05); the decision landed on CatBoost MultiQuantile at alpha **0.05**, **0.5** and **0.85**, which produced asymmetric **80 percent** prediction intervals that hug the ground floor and cap the upper **15 percent** against summer ghost spikes.
- Made the September evaluation robust with a **3-pass** outlier system: pH narrowed from **0–14** to **7.0–9.0**, Oct/Nov caps of **15.0** for chlorophyll and **50.0** for turbidity, a rolling z-score over **48 hours**, and 5 sparse sensors plus 3 concurrent proxies (phycocyanin_abs, phycocyanin_abs_comp, toc) excluded — surfacing the honest R² split of **0.86** for dissolved oxygen and **0.81** for pH.
- Reconstructed Oct/Nov gaps with IterativeImputer MICE, ran a full memory-buffer recomputation across all 6 lag features, generated a synthetic winter canvas at **4 °C** floor and **0.4** diurnal amplitude, and embedded it all in an orchestrator with gate checks, ecological clips (dissolved oxygen **4.0–18.0**, pH **6.0–9.0**) and a **0.003** pH-per-hour velocity clamp.

Technologies: Python, CatBoost, Prophet, scikit-learn, MICE

Aug 2023 – Aug 2024
India

SS Engineers and Contractors

Junior Associate Software Developer

- Ran the internal data dashboards, analytics platforms and employee portals used company-wide, tasked with building and maintaining the frontend features the teams relied on daily, contributing React UI components across all internal products that stayed in daily use throughout the year.
- With a legacy AngularJS application needing to run alongside newer React micro-frontends in an existing Module Federation setup, tasked with porting the client's screens without breaking the live product, ported around **8 routes** from AngularJS to React inside the Module Federation shell, shipped over **4 months** in incremental releases with no production outages during rollout.

Technologies: React, module federation, Playwright, AngularJS, HTML5, CSS3

Feb 2023 – Jul 2023
India

SS Engineers and Contractors

Front End Developer Intern

- During a six-month internship, tasked with learning the codebase and then taking on smaller UI work under senior review across the internal data dashboards and employee portals; worked closely with senior developers through the codebase and then delivered focused UI components such as charts, filters and profile pages, each iterated on code-review feedback until approved.

Technologies: React, HTML5, CSS3, Git, code review workflow

P E R S O N A L P R O J E C T S

2025

Multi-Agent RAG with LLM-as-Judge and multilingual EN and DE support

- For an English-only Hybrid RAG policy analysis system with BM25 and dense vector search over a **14-document** policy corpus that could not serve German-speaking users, tasked with adding multilingual support without duplicating the corpus, migrated embeddings and retrieval to a shared paraphrase-multilingual-MiniLM-L12-v2 vector space so a German query retrieves English sources and is answered end-to-end in German.
- Because every agent in the graph re-ran language detection and produced inconsistent output languages, tasked with creating a single source of truth, built a LanguageAgent that centralises seeded language detection with a confidence floor, passes the output-language directive to every downstream agent, and routes preprocessing to language-appropriate spaCy pipelines with an empty multilingual fallback, logging language per chunk so retrieval and evaluation can be sliced by language.
- To measure answer quality without manual scoring, implemented an LLM-as-Judge JudgeAgent that scores answers **1 to 5 on 5 dimensions** (Groundedness, Relevance, Completeness, Citation Quality, Language Quality) in JSON mode at temperature 0, eliminating self-preference bias by running the judge on a different local model (Qwen **2.5 14B**) than the generator (Mistral **7B**), with a self_judged flag on every report and hard failure handling so a missing judge model does not silently regress to self-judging.

Technologies: Python, LangGraph Multi Agent, Ollama with Mistral 7B and Qwen 2.5 14B, Pinecone, paraphrase-multilingual-MiniLM-L12-v2, spaCy, BM25

E D U C A T I O N

Apr 2025 – present
Heidelberg

M.Sc. Data Science and Analytics
SRH University of Applied Sciences Heidelberg, GPA 1.9

2019 – 2023
Greater Noida, India

Bachelor of Technology in Computer Science
GL Bajaj Institute of Technology and Management, CGPA 7.3 of 10

R E S E A R C H A N D T H E S I S

2022 – 2023
Greater Noida, India

Bachelor thesis: Diabetes prediction with Machine Learning
GL Bajaj Institute of Technology and Management, IEEE style paper

- For a bachelor thesis on diabetes prediction on a small clinical dataset of **768 patients**, tasked with delivering a robust model comparison the examiners could verify, built a complete end-to-end machine learning pipeline with six classifiers, **10-fold** cross-validation and per-model confusion matrices, yielding a model comparison that held up in the defence.
- After spotting biologically impossible zero values in the source data that the original authors had overlooked, tasked with restoring data integrity before any model training, applied IQR-based outlier handling and clean imputation, taking the dataset from silently faulty to a clean training foundation for every downstream model.
- With a 65-to-**35-class** imbalance that made accuracy a deceptively flattering headline metric, tasked with choosing an evaluation that does not hide minority-class errors, switched the headline metric to ROC AUC, which exposed real error patterns that the accuracy number had masked and gave the work an honest performance comparison.
- As the results were publication-ready in substance and not just submittable, tasked with writing them up formally, wrote an IEEE paper with an honest section on limitations and improvement suggestions for a follow-up study, which the supervisor accepted as publication-ready in substance.

Technologies: Python, scikit-learn, Pandas, Seaborn

C E R T I F I C A T E S

- NVIDIA: Building LLM Applications With Prompt Engineering, issued November 2025.
- AWS Academy Graduate: AWS Academy Cloud Foundations, issued July 2025.
- Google Data Analytics: Foundations, Data, Data, Everywhere, issued April 2025 on Coursera.

A W A R D S

- USAII Global AI Hackathon 2026: Finalist at Graduate Level, recognised by the United States Artificial Intelligence Institute for innovation, technical creativity and applied AI on real-world challenges.

L A N G U A G E S

English Fluent, written and spoken
German B1, progressing toward B2
Hindi Native language